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Department of Industrial Engineering & Management

- + ENMG 616 – Advanced Optimization Techniques & Algorithms
- + Research Project: Fairness in Machine Learning
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“Fairness Beyond Disparate Treatment and Disparate Impact: Learning Classification without Disparate Mistreatment”

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OUTLINE

- + 1- Introduction & Motivation
- + 2- Literature Review
- + 3- Problem Formulation & Methodology
- + 4- Experiments & Results
- + 5- Relation to the Class Material

1- Introduction & Motivation

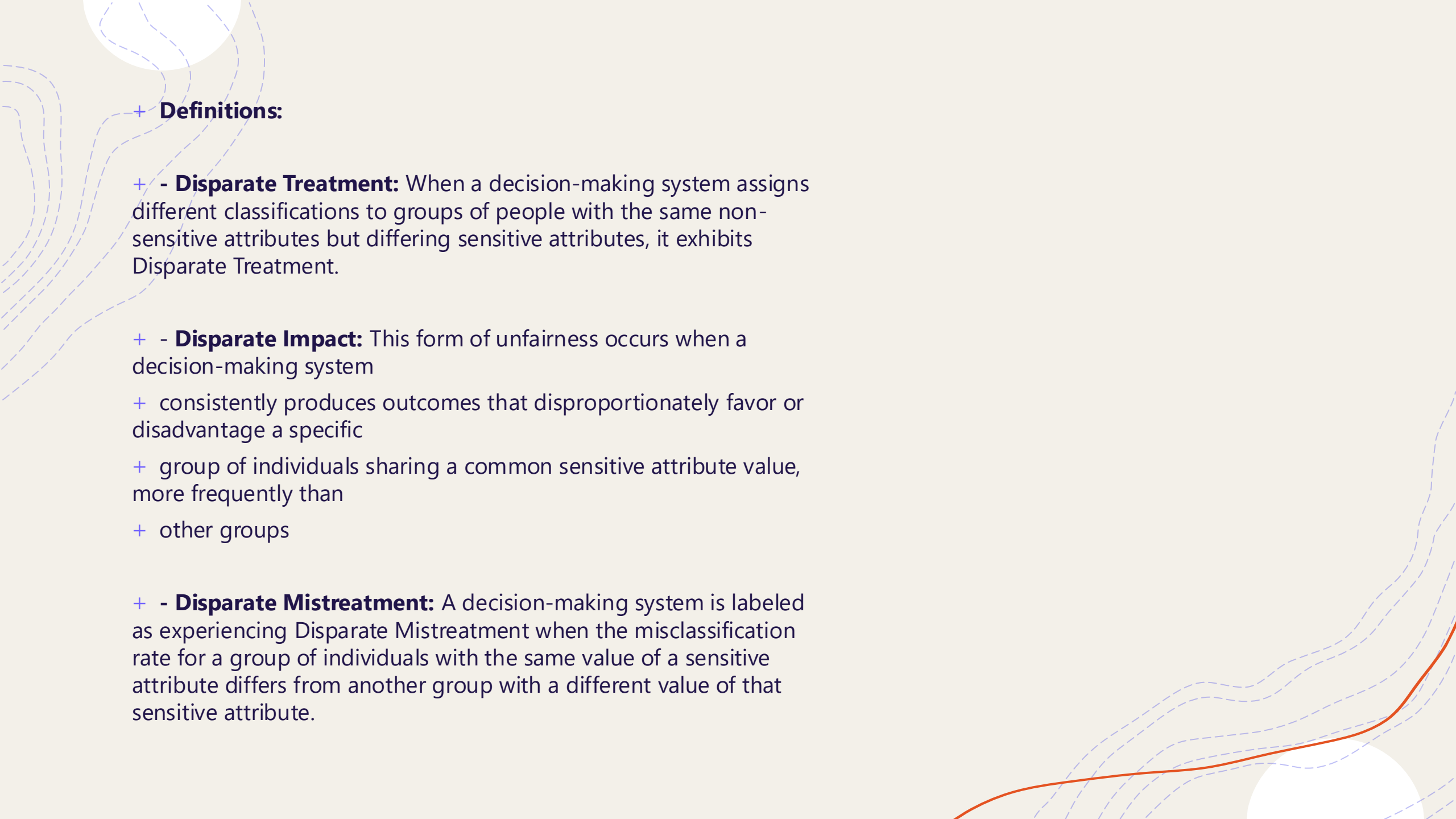
- Huge increase in Automated data-driven decision systems

- Concerns about unfairness towards groups of people with sensitive attributes such as gender or race.

- Several laws tackled Disparate Treatment & Disparate Impact, when ground truth is not provided.

- If the ground truth is provided, we can find that sometimes a certain group of people should receive different decisions.

→ **DISPARATE MISTREATMENT is introduced**



+ **Definitions:**

+ - **Disparate Treatment:** When a decision-making system assigns different classifications to groups of people with the same non-sensitive attributes but differing sensitive attributes, it exhibits Disparate Treatment.

+ - **Disparate Impact:** This form of unfairness occurs when a decision-making system

+ consistently produces outcomes that disproportionately favor or disadvantage a specific

+ group of individuals sharing a common sensitive attribute value, more frequently than

+ other groups

+ - **Disparate Mistreatment:** A decision-making system is labeled as experiencing Disparate Mistreatment when the misclassification rate for a group of individuals with the same value of a sensitive attribute differs from another group with a different value of that sensitive attribute.

Example:

We have 3 classifiers that will decide whether or not to stop pedestrians because they think they might have weapons. We have one sensitive attribute that is Gender, and two non sensitive attributes, Clothing Bulge and Proximity to a crime scene.

Gender	Clothing Bulge	Proxy to crime scene	TRUTH	C1	C2	C3
Male 1	1	1	YES	1	1	1
Male 2	1	0	YES	1	1	0
Male 3	0	1	NO	1	0	1
Female 1	1	1	YES	1	0	1
Female 2	1	0	NO	1	1	1
Female 3	0	0	YES	0	1	0

Gender	False Positive Rate	False Negative Rate
Males	C1 - 1.0 C2 - 0 C3 - 1.0	C1 - 0.0 C2 - 0.0 C3 - 0.5
Females	C1 - 1.0 C2 - 1 C3 - 1.0	C1 - 0.5 C2 - 0.5 C3 - 0.5

	% of males stopped	% of females stopped
C1	3/3 = 100%	2/3 = 67%
C2	2/3 = 67%	2/3 = 67%
C3	2/3 = 67%	2/3 = 67%

	Disparate Treatment	Disparate Impact	Disparate Mistreatment
C1	NO - Males 1&2 Females 1&2 have the same non-sensitive attributes and received the same decision	YES - 100% of the males got stopped and 67% of females got stopped	YES - FNR (females) > FNR (males)
C2	YES - Male 2 & Female 2 have same non sensitive attributes and Male 2 got stopped but not Female 2	NO - 67% of males and 67% of females got stopped. Equal Proportions	YES - FNR (females) > FNR (males) FPR (females) > FPR (males)
C3	YES - Male 2 & Female 2 have same non sensitive attributes and Female 2 got stopped but not Male 2	NO - 67% of males and 67% of females got stopped. Equal Proportions	NO - Same FPR and FNR for males and females

2- Literature Review

- Many Studies only tackle Disparate Treatment & Impact

- Goel et al: Raised awareness about unfairness towards black people regarding racial disparities in the detection of weapons with civilians.

- Hardt et al: tackle disparate mistreatment: Ad-

just the unfair system's probability estimates after it makes predictions. This adjustment

helps the system set different decision thresholds for various sensitive attribute groups.

However, this method has a limitation: it needs sensitive attribute information at the time

of decision-making. So, it may not work when this information is unavailable.

3- Problem Formulation & Methodology

Min $L(\theta)$

- + We aim to find a function $f(x)$ that maps user feature vectors $x \in R^d$ and class labels $y \in \{-1, 1\}$.
- + To achieve this, we need to determine a decision boundary $\theta^* = \operatorname{argmin} L(\theta)$ (a surface separating the data-points of different classes) using a training dataset $D = \{(x_i, y_i)\}$ for $i = 1$ to N .
- + Then, when we're given a new feature vector x , the system will be able to predict whether $\hat{y} = 1$ (if $d\theta^*(x) \geq 0$) or $\hat{y} = -1$ if $(d\theta^*(x)) < 0$.
- + $d\theta^*(x)$ represents the signed distance from x to the decision boundary. If it is on the positive side, $\hat{y} = 1$, and $\hat{y} = -1$ otherwise.
- + In this paper, a logistic regression model will be used. The decision boundary here will be the hyperplane defined by $\theta^T \mathbf{x} = 0$, and $d\theta(x) = \theta^T \mathbf{x}$. Consider that we have a feature vector x and a sensitive attribute $z = \{0, 1\}$. The constraints for the three notions of fairness will be defined as follow:

Fairness Constraints

Disparate Treatment: The Probability that a classifier outputs a value of $\hat{y}$, given the feature vector $\mathbf{x}$, must remain the same if the sensitive attribute z is also given

$$[P(\hat{y}|\mathbf{x}, z) = P(\hat{y}|\mathbf{x})]$$

Disparate Impact: The Probability of getting the output $\hat{y}=1$ (or 0) should be the same for both values of the sensitive feature z

$$[P(\hat{y} = 1|z = 0) = P(\hat{y} = 1|z = 1)]$$

Disparate Mistreatment: The misclassification rates for different groups of people having different values of z should remain the same. Here we have many misclassification measures, such as:

- Overall Misclassification Rate:

$$[P(\hat{y} \neq y|z = 0) = P(\hat{y} \neq y|z = 1)]$$

- False Positive Rate:

$$[P(\hat{y} \neq y|z = 0, y = -1) = P(\hat{y} \neq y|z = 1, y = -1)]$$

- False Negative Rate:

$$[P(\hat{y} \neq y|z = 0, y = 1) = P(\hat{y} \neq y|z = 1, y = 1)]$$

We will only cover these types, but there's also False Omission rate and False Discovery Rate

Simplifying the fairness constraints to make them a Discipline Convex Concave Program (DCCP)

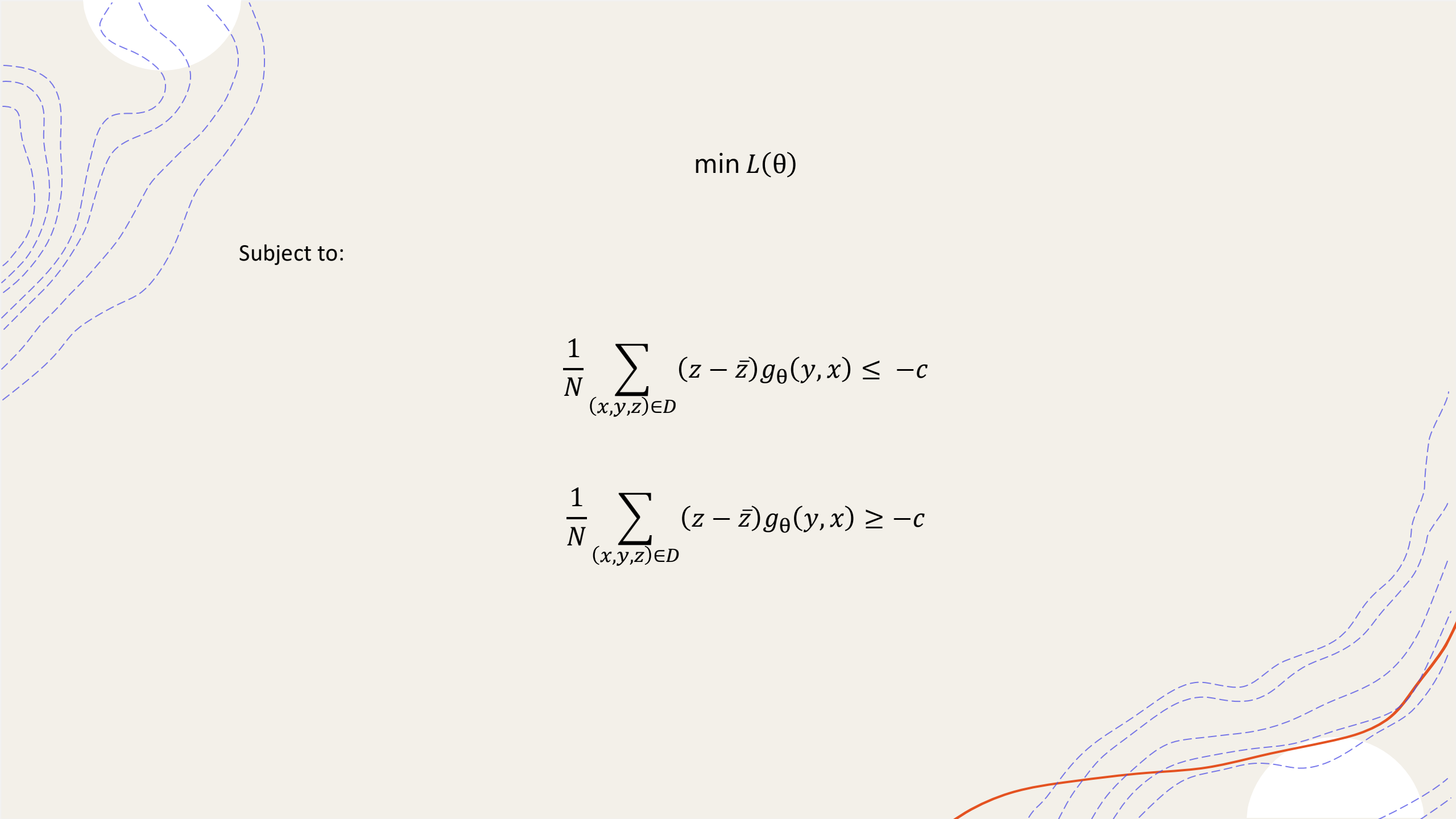
- Step 1: Measure disparate mistreatment using the covariance between the users' sensitive attribute values (z), and the signed distance between the feature vectors of misclassified users and the decision boundary $g(y, x)$

$$\text{Cov}(z, g_{\theta}(y, x)) = E[(z - \bar{z})(g_{\theta}(y, x) - \overline{g_{\theta}}(y, x))] \approx \frac{1}{N} \sum_{(x, y, z) \in D} (z - \bar{z}) g_{\theta}(y, x)$$

$$\textbf{OMR: } g_{\theta}(y, x) = \min(0, y \cdot d_{\theta}(x))$$

$$\textbf{FPR: } g_{\theta}(y, x) = \min\left(0, \frac{1-y}{2} \cdot y d_{\theta}(x)\right)$$

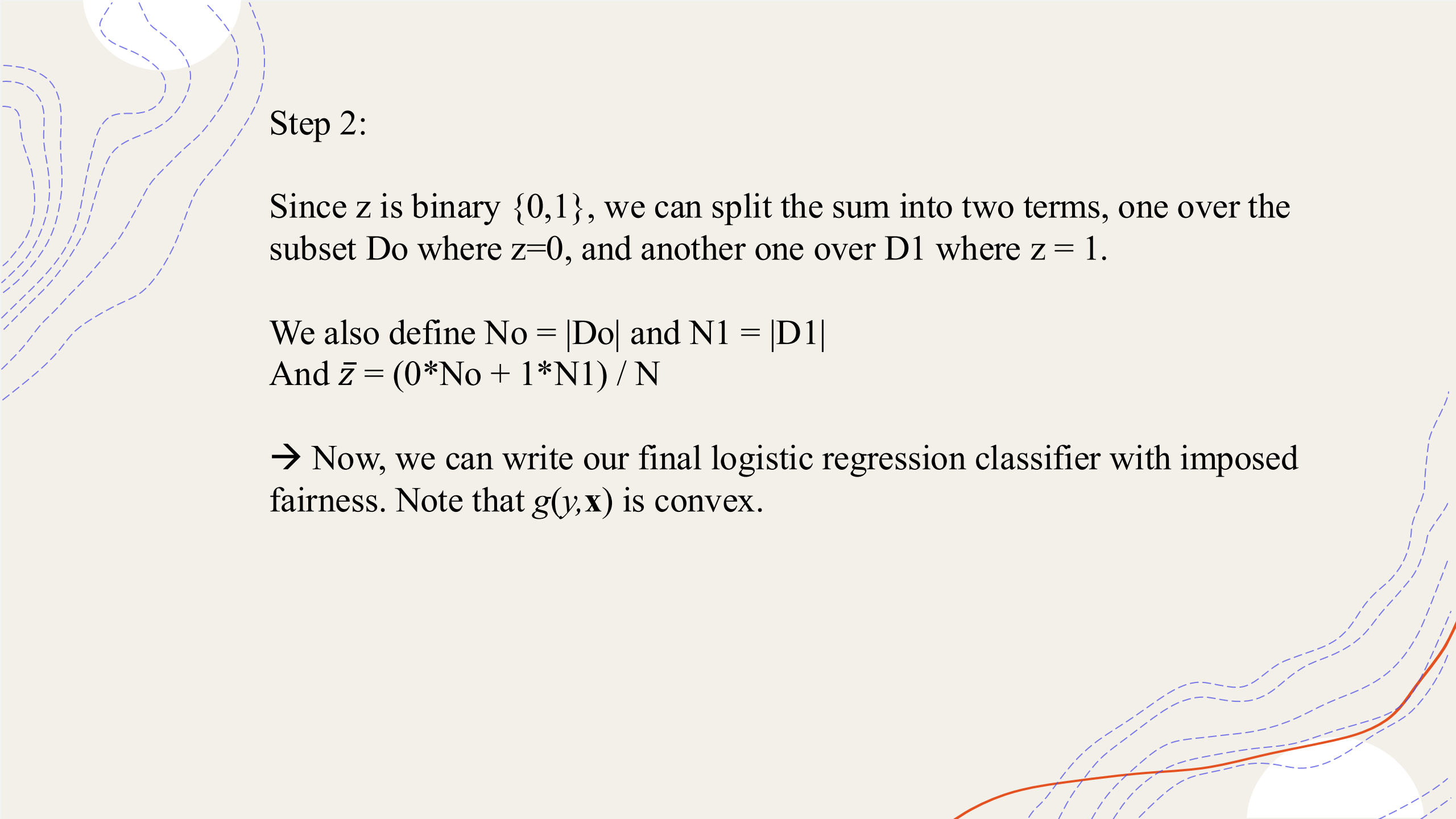
$$\textbf{FNR: } g_{\theta}(y, x) = \min\left(0, \frac{1+y}{2} \cdot y d_{\theta}(x)\right)$$


$$\min L(\theta)$$

Subject to:

$$\frac{1}{N} \sum_{(x,y,z) \in D} (z - \bar{z}) g_{\theta}(y, x) \leq -c$$

$$\frac{1}{N} \sum_{(x,y,z) \in D} (z - \bar{z}) g_{\theta}(y, x) \geq -c$$



Step 2:

Since z is binary $\{0,1\}$, we can split the sum into two terms, one over the subset D_0 where $z=0$, and another one over D_1 where $z = 1$.

We also define $N_0 = |D_0|$ and $N_1 = |D_1|$

And $\bar{z} = (0 \cdot N_0 + 1 \cdot N_1) / N$

→ Now, we can write our final logistic regression classifier with imposed fairness. Note that $g(y, \mathbf{x})$ is convex.

$$\text{minimize } - \sum_{(x,y) \in D} \log p(y|x; \theta)$$

Subjected to:

$$-(N_1/N) \sum_{(x,y) \in D_0} g_\theta(y,x) + (N_0/N) \sum_{(x,y) \in D_1} g_\theta(y,x) \leq c$$

$$-(N_1/N) \sum_{(x,y) \in D_0} g_\theta(y,x) + (N_0/N) \sum_{(x,y) \in D_1} g_\theta(y,x) \geq -c$$

DCCP can be solved by using advances in convex-concave programming.

4- Experiments and Results

Synthetic Dataset

$$D(\text{FPR}) = [P(\hat{y} \neq y | z = 0, y = -1) - P(\hat{y} \neq y | z = 1, y = -1)]$$

$$D(\text{FNR}) = [P(\hat{y} \neq y | z = 0, y = 1) - P(\hat{y} \neq y | z = 1, y = 1)]$$

At each experiment, 4 samples (each 2500 points) with different distributions will be generated, and the logistic classifier will be solved first without the fairness constraints to see the new results when imposing them.

1- Removing disparate mistreatment on FPR or FNR		
	Unconstrained Classifier	Fairness constraint on FPR
Accuracy	0.85	0.8
D (FPR)	0.19	0.02
D (FNR)	0	0

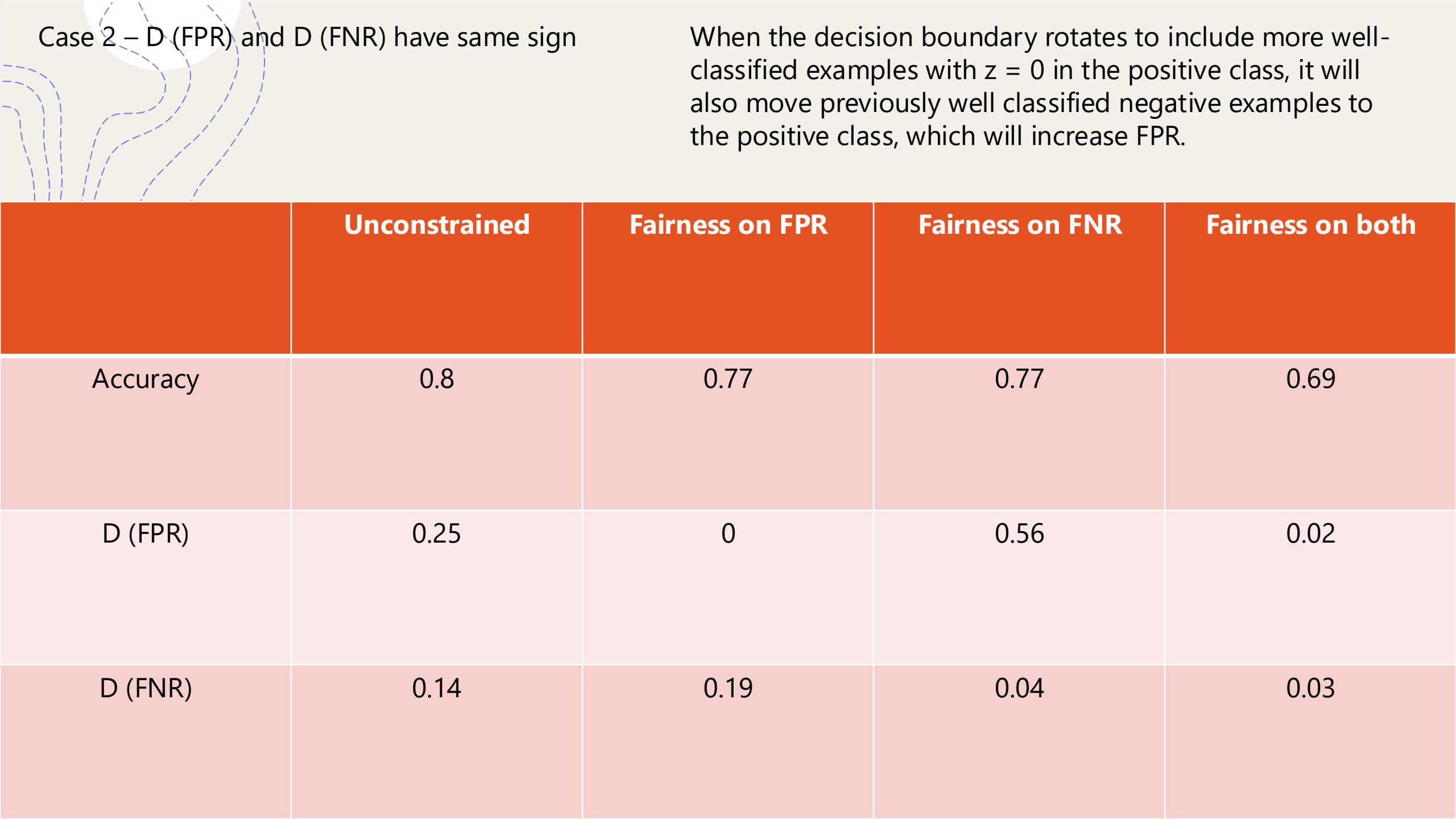
2) Removing Disparate mistreatment on both FPR & FNR

Case 1 – D (FPR) & D(FNR) have opposite signs

When the decision boundary rotated due to the constraints, misclassified examples with $z = 1$ move to the negative class as well as well classified examples with $z = 1$. This caused a decrease in FPR and an increase on FNR

	Unconstrained	Fairness on FPR	Fairness on FNR	Fairness on both
Accuracy	0.78	0.75	0.75	0.75
D (FPR)	-0.16	0	0	0
D (FNR)	0.19	0	0	0

	Unconstrained	Fairness on FPR	Fairness on FNR	Fairness on both
Accuracy	0.78	0.75	0.75	0.75
D (FPR)	-0.16	0	0	0
D (FNR)	0.19	0	0	0





Performance Evaluation on a real dataset

- + We will compare:
- + - Method 1 removing only disparate mistreatment
- + - Hardt et al. Method
- + - Baseline: Remove disparate mistreatment by:
 - + Running the unfair classifier, selecting the set of misclassified data with higher error, and re-training the model with an increased penalty on this set until fairness is achieved.

Real dataset info

- Info on offenders' demographic features.

- Take into account race (black, white).

- Use the ground truth to build different logistic classifiers

	FPR			FNR			BOTH CONSTRAINTS		
	Acc	D(FPR)	D(FNR)	Acc	D(FPR)	D(FNR)	Acc	D(FPR)	D(FNR)
Unfair	0.668	0.18	-0.3						
NO FPR	NO FPR CNT	NO FPR CNT	NO FPR CNT						
Method 1	0.66	0.06	-0.14	0.662	0.03	-0.1	0.661	0.03	-0.11
Baseline	0.64	0.03	-0.11	0.66	0.00	-0.07	0.660	0.01	-0.09
Hardt	0.66	0.02	-0.08	0.65	-0.06	-0.01	0.645	-0.01	-0.01

Results

Due to the small size, disparate mist isn't removed completely.

+ For all three methods, controlling for disparate mistreatment on false positive rate (false negative rate) also helps decrease disparate mistreatment on false negative rate (false positive rate). Moreover, all three methods are able to achieve similar accuracy for a given level of fairness

Relation to class material

- Constrained
Optimization

- Convexity

- Logistic
Regression